



दक्षिण बिहार केंद्रीय विश्वविद्यालय

CENTRAL UNIVERSITY OF SOUTH BIHAR

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Question Paper of End-Term Examination Dec 2025

Name of Department : Department of Computer Science Programme : MSc. (AI)
Session : 2024-2026 Semester : III
Course Title : Soft Computing Course Code : CAI91DE02004
Course Credit : 4 Max. Marks : 70 Duration : 3 hours
Enrolment No. (to be filled by Student) : CUS87401332028

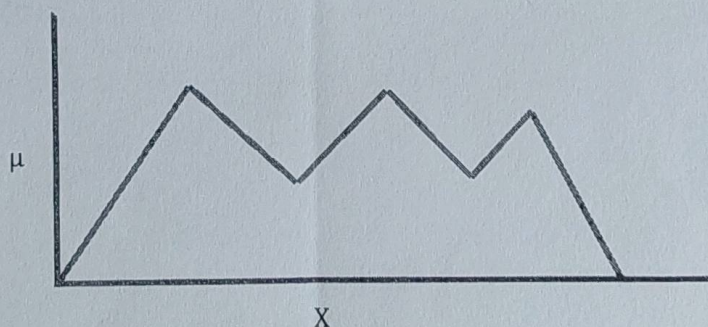
There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis.

Section – A

Answer ALL the questions in approximately 50 words. Each question carries 1 Points.

(10 x 1 = 10 Points)

- Q1. Draw a diagram to state the notion of computing?
Q2. Why the soft computing gives stochastic solution where as hard computing gives deterministic solution?
Q3. What are the soft computing concepts involved in the hand written character recognition system?
Q4. Use your intuition to draw a fuzzy membership function for medium height persons.
Q5. What do you mean by fuzzification?
Q6. Why is the following fuzzy set non convex?



Q7. Show that

$$(\tilde{A} \cap \tilde{B})_{\alpha} = A_{\alpha} \cap B_{\alpha}$$

where A_{α} represents alpha-cut.

Q8. How is the supervised learning different from unsupervised learning?

Q9. Draw a diagram of a single artificial neuron along with its components.

Q10. What is the need of genetic algorithm?

Section – B

Answer ANY FIVE questions in approximately 300 words. Each question carries 6 Points.

(5 x 6 = 30 Points)

Q1. Justify that the following is a valid definition for the fuzzy set?

$$\mu(x) = \text{Degree}(x \in A)$$

where $\mu: X \rightarrow [\text{Membership Degrees}]$

Then the fuzzy set A can be given by

$$A = \{(x, \mu(x))\}, x \in X$$

Q2. With help of graphs, explain intersection operations in fuzzy sets. Also find union, intersection and complement operations in the following fuzzy sets. [3+3]

$$\bar{A} = \left\{ \frac{0.1}{x} + \frac{0.5}{y} + \frac{0.4}{z} + \frac{0.2}{w} \right\}$$

$$\bar{B} = \left\{ \frac{0.2}{x} + \frac{0.5}{y} + \frac{0.4}{z} + \frac{0.7}{w} \right\}$$

Q3. Explain the following fuzzy defuzzification methods:-

I. Centroid method

II. Centre of Sum method

Q4. Describe the role of the loss function and gradient descent in backpropagation neural network.

Q5. Design a MP-Neuron for implementation of ANDNOT function.

Q6. Discuss three parent crossover with the help of suitable example.

Section – C

Answer ANY TWO questions in approximately 1000 words. Each question carries 15 Points.

(2 x 15 = 30 Points)

Q1. Define alpha cut. Write the properties of the alpha cut. Use the definition to show the following for the fuzzy set A: [3+4+4+4]

I. $\text{Core}(A) = A_1$

II. $A_0 = X$

Q2. Discuss the following fuzzy membership function along with a neat diagram for each: [5+5+5]

I. Triangular Membership function

II. Trapezoidal Membership function

III. Gaussian Membership function

Q3. Discuss the different encoding mechanisms used in the genetic algorithm.